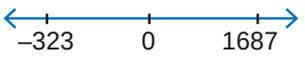
Example 1.30

How many years are between 323 BC(BCE) and 1687 AD(CE)?

Solution

Years in AD(CE) are taken as positive integers and BC(BCE) as negative integers.

Therefore, the difference is



= 1687 – (–323)

= 1687 + 323 = 2010 years

Exercise 1.5

1. One night in Kashmir, the temperature is -5°C. Next day the temperature is 9°C. What is the increase in temperature?
2. An atom can contain protons which have a positive charge $(+)$ and electrons which have a negative charge $(-)$. When an electron and a proton pair up, they become neutral (0) and cancel the charge out. Now, Determine the net charge:
   (i) 5 electrons and 3 protons $\rightarrow$ $-5+3=-2$ that is 2 electrons $\ominus\ominus$
   (ii) 6 Protons and 6 electrons $\rightarrow$
   (iii) 9 protons and 12 electrons $\rightarrow$
   (iv) 4 protons and 8 electrons $\rightarrow$
   (v) 7 protons and 6 electrons $\rightarrow$
3. Scientists use the Kelvin Scale (K) as an alternative temperature scale to degrees Celsius (°C) by the relation $T^{\circ}C=(T+273)$ K.
   Convert the following to kelvin:
   (i) -275° C (ii) 45° C (iii) -400° C (iv) -273° C
4. Find the amount that is left in the student's bank account, if he has made the following transaction in a month. His initial balance is ₹ 690.
   (i) Deposit $(+)$ of ₹ 485
   (ii) Withdrawal $(-)$ of ₹ 500
   (iii) Withdrawal $(-)$ of ₹ 350
   (iv) Deposit $(+)$ of ₹ 89
   (v) If another ₹ 300 was withdrawn, what would the balance be?
5. A poet Tamizh Nambi lost 35 pages of his 'lyrics' when his file had got wet in the rain. Use integers, to determine the following:
   (i) If Tamizh Nambi wrote 5 page per day, how many day's work did he lose?
   (ii) If four pages contained 1800 characters, (letters) how many characters were lost?
   (iii) If Tamizh Nambi is paid ₹250 for each page produced, how much money did he lose?

7th_Maths_T1_EM_Chp1.indd 28 12/26/2019 4:24:00 PM